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MIN et al.(10) **Pub. No.: US 2025/0282023 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **POLISHING PAD CONDITIONER POSITION
DETECTION MODULE, POLISHING
DEVICE, AND POLISHING DETECTION
METHOD USING THE POSITION
DETECTION MODULE****Publication Classification**

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Keunju Ha, Suwon-si (KR)(21) Appl. No.: **18/979,730**(22) Filed: **Dec. 13, 2024**(30) **Foreign Application Priority Data**

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(57) **ABSTRACT**

A polishing device includes a rotatable platen; a polishing pad placed on an upper surface of the platen and having a polishing surface; a polishing head disposed above the polishing pad and configured for mounting a substrate on a bottom surface of the polishing head; a driver configured to rotate the platen with respect to an axis perpendicular to the polishing surface of the polishing pad; a slurry supply outlet configured to supply slurry to the polishing surface of the polishing pad; a polishing pad conditioner for conditioning the polishing surface of the polishing pad; and a polishing pad conditioner position detection module configured to detect whether a disk of the polishing pad conditioner is correctly aligned.

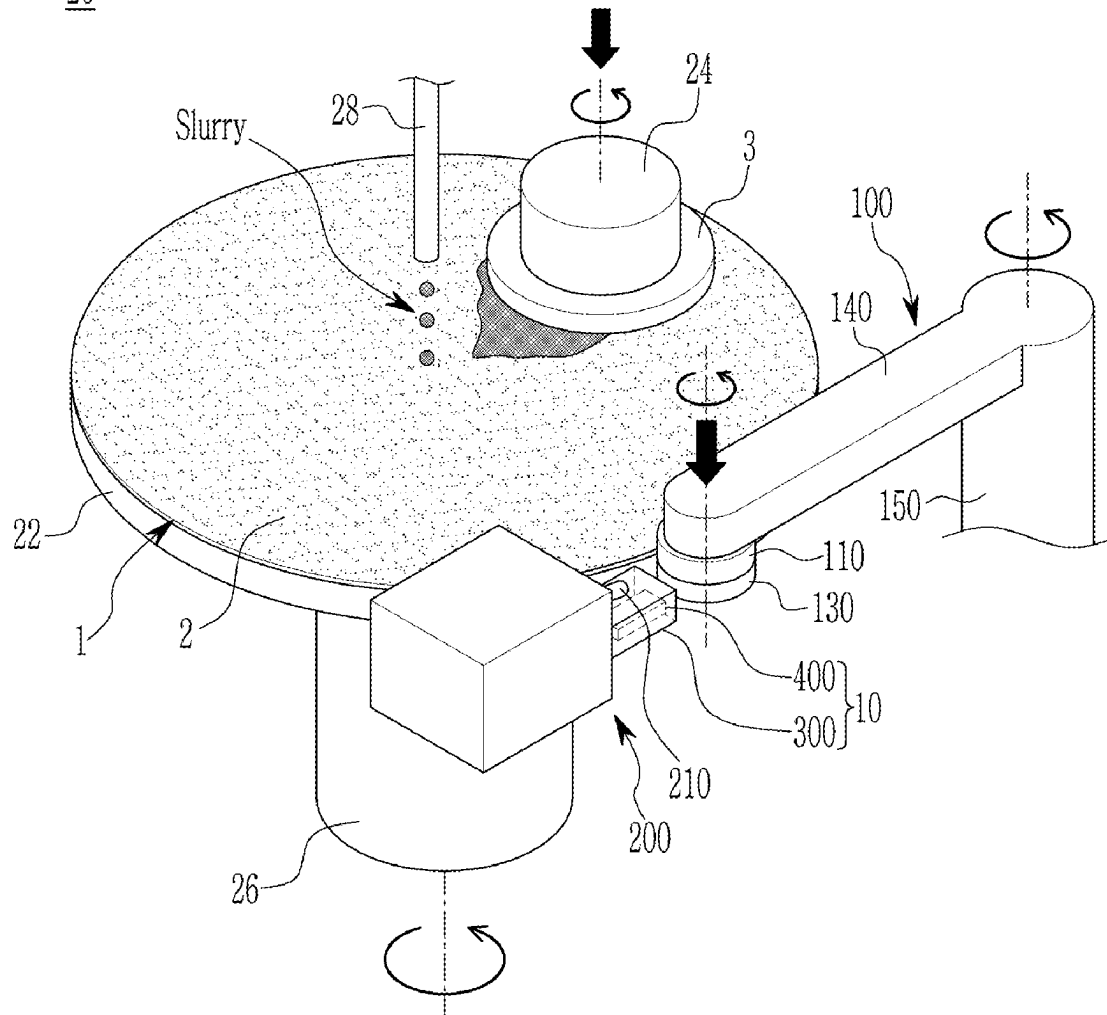
20

FIG. 1

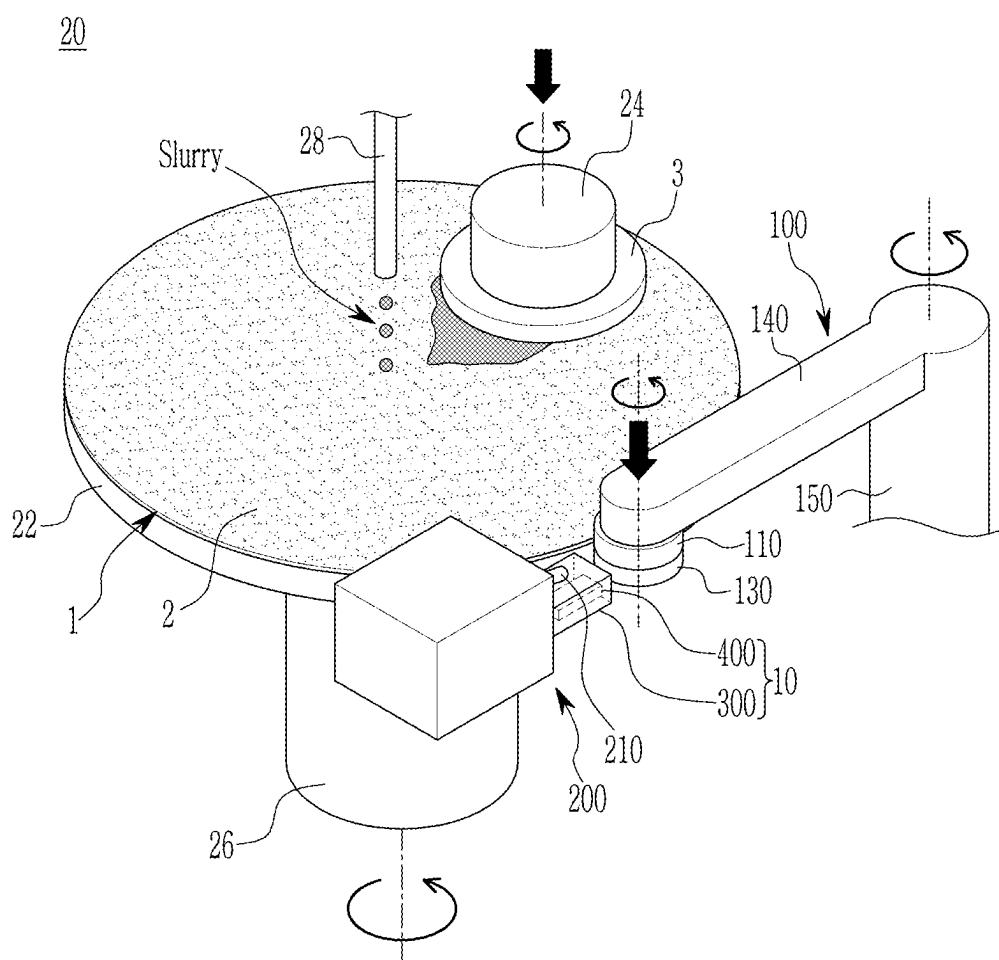


FIG. 2

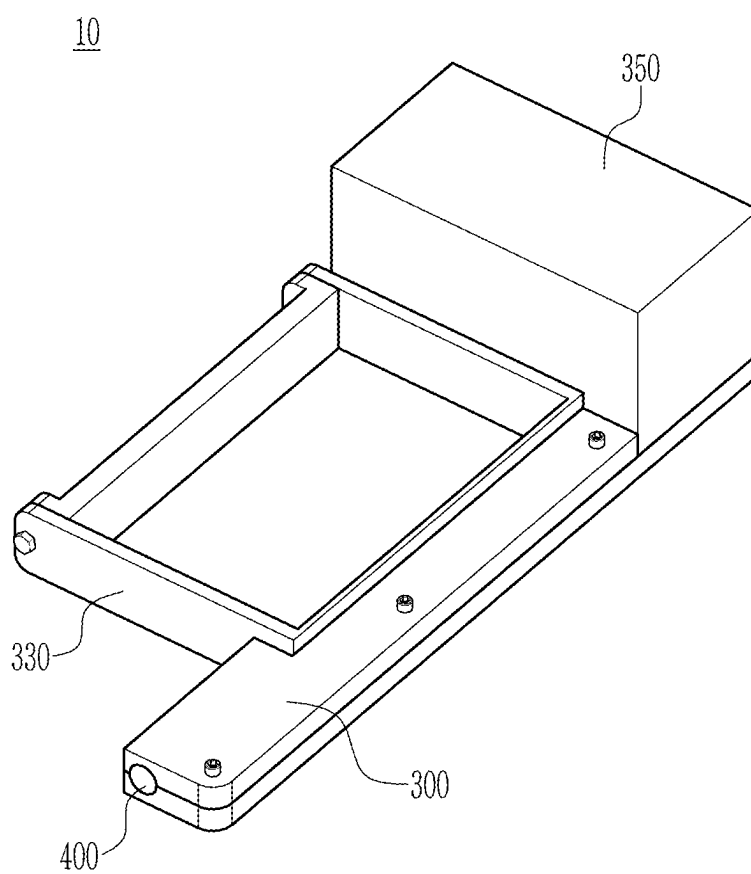


FIG. 3

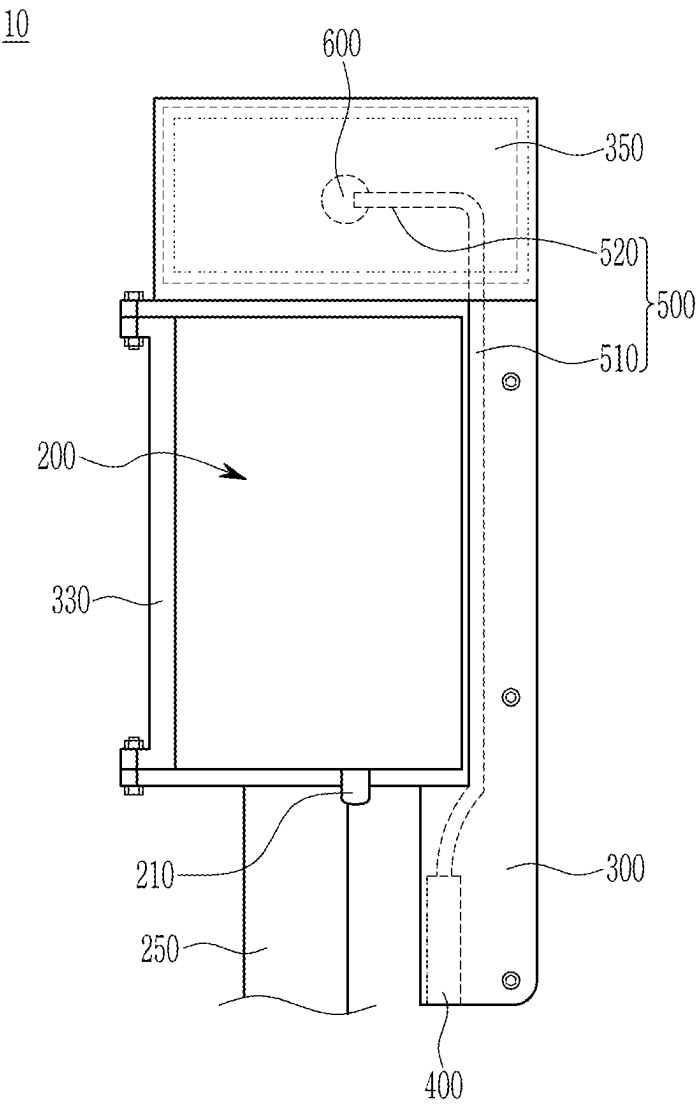


FIG. 4

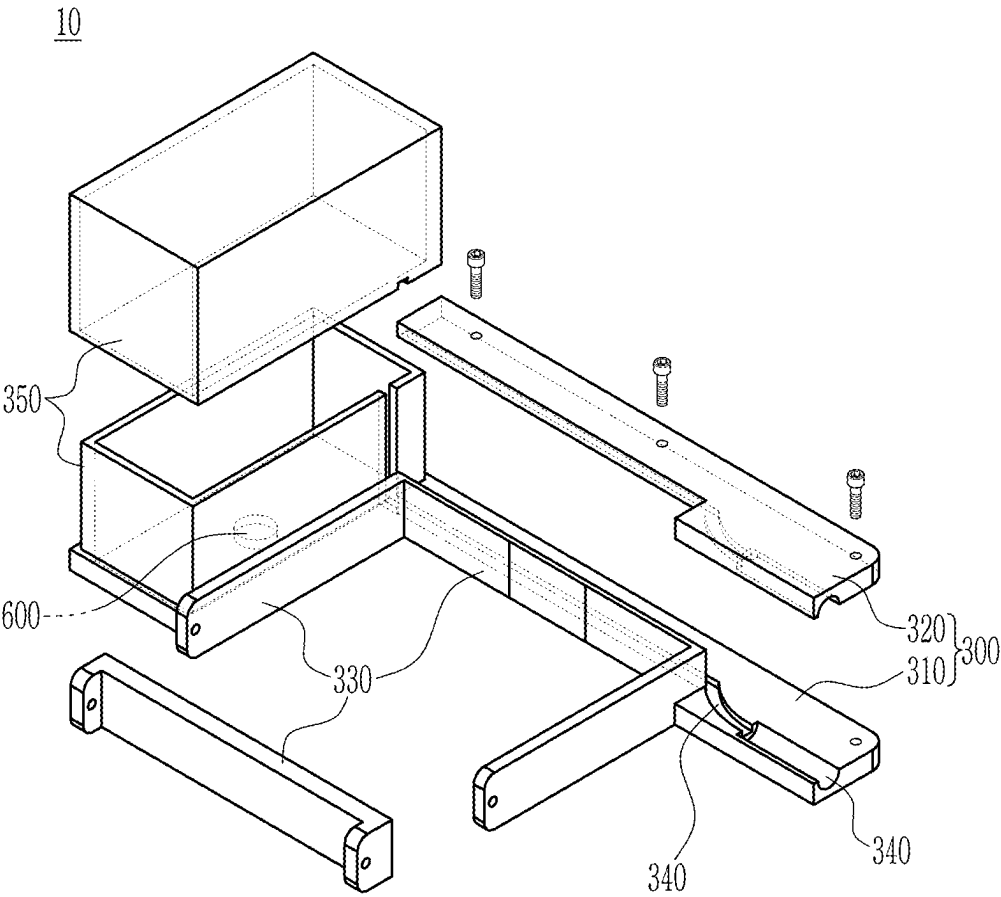


FIG. 5

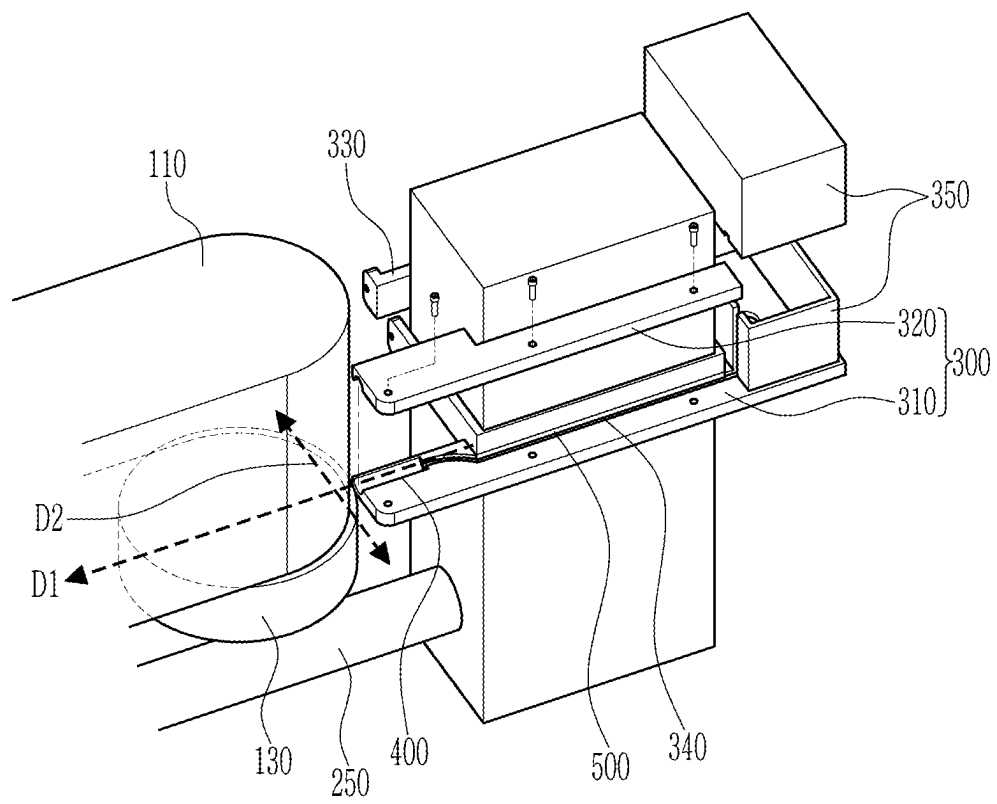


FIG. 6A

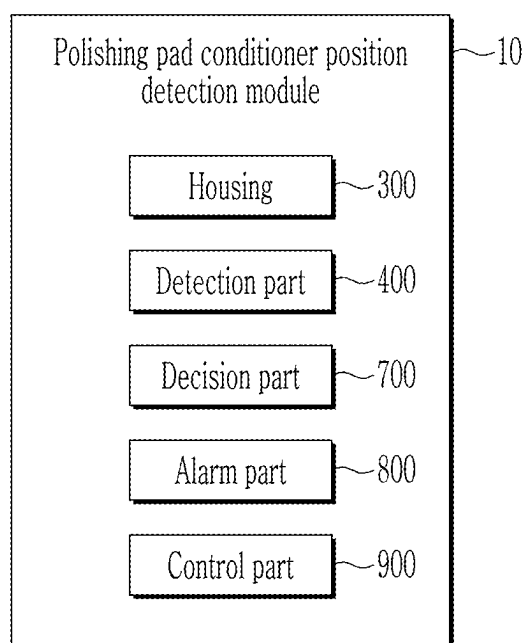


FIG. 6B

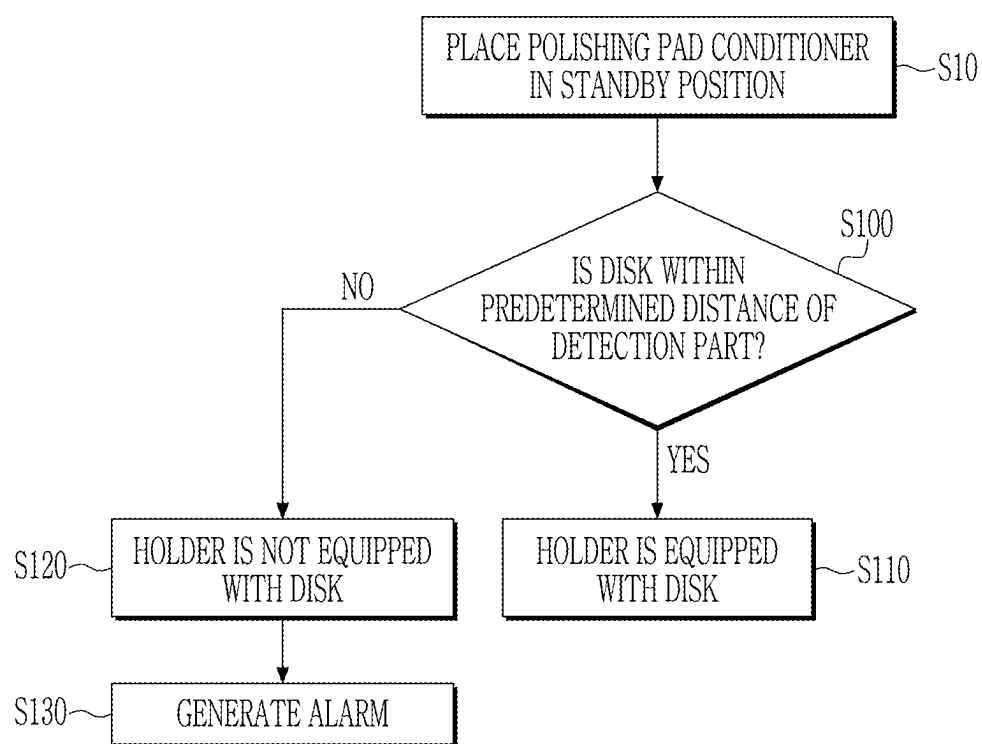


FIG. 6C

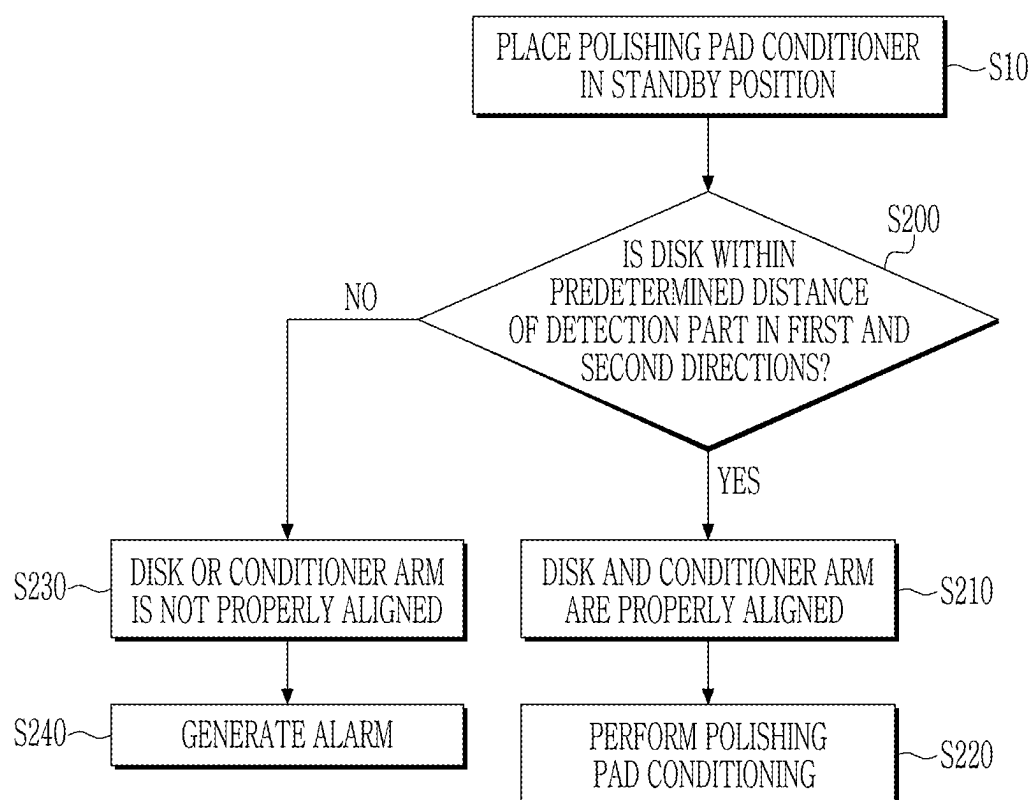


FIG. 7

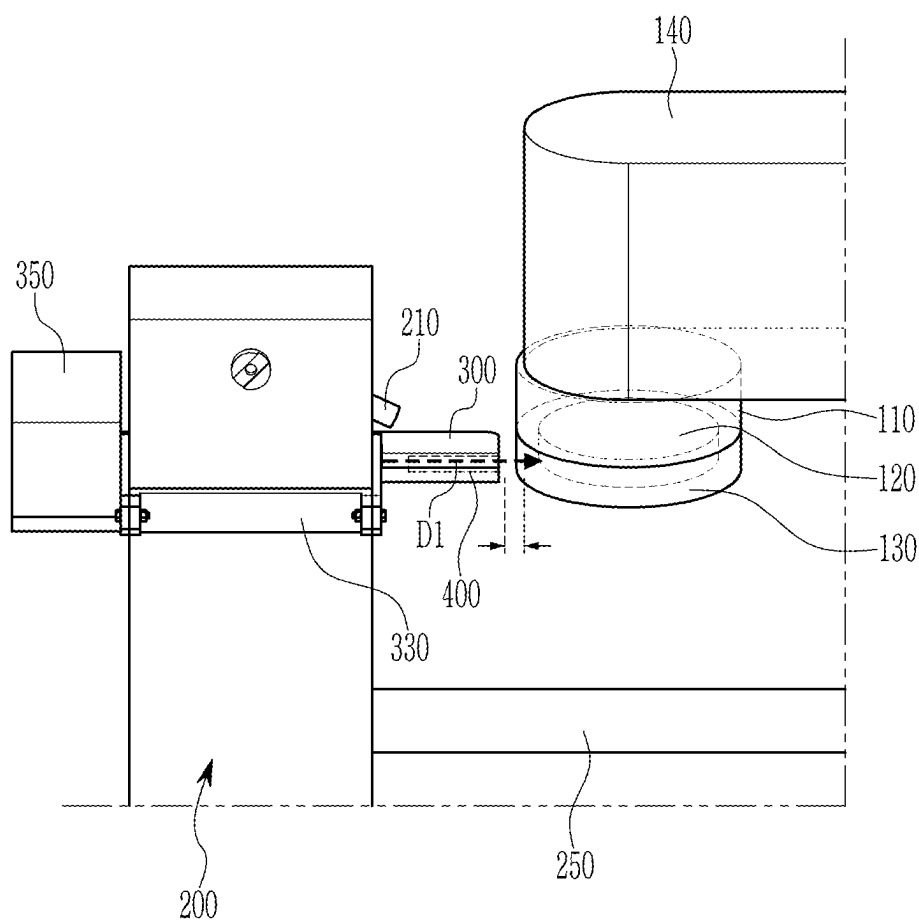


FIG. 8

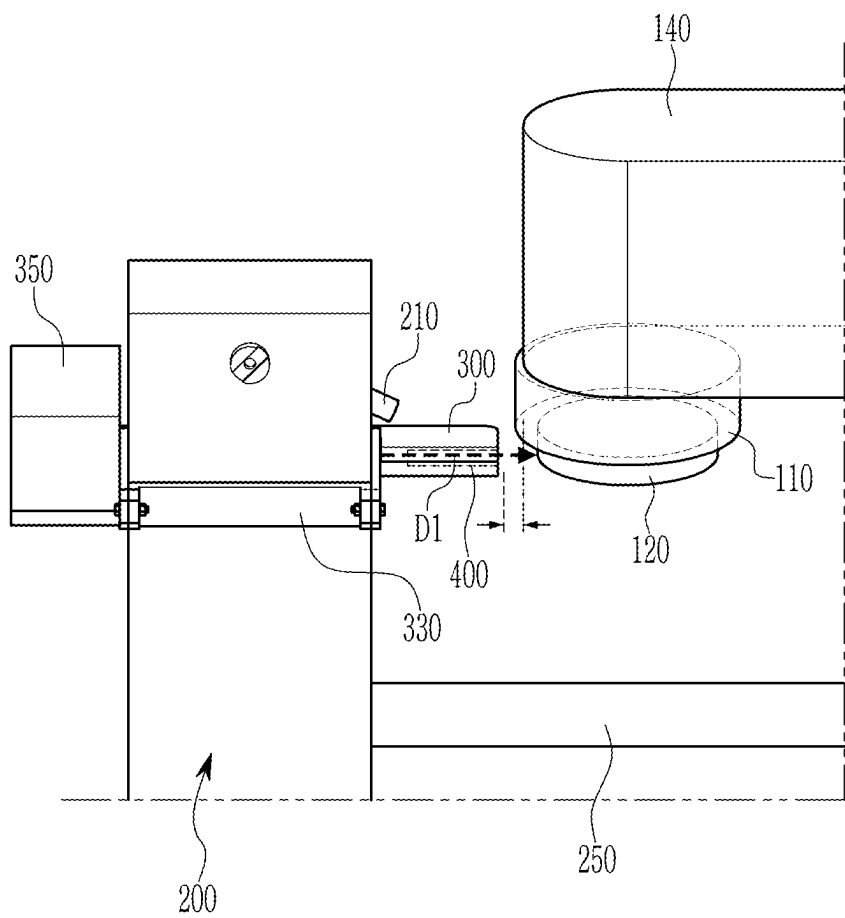


FIG. 9

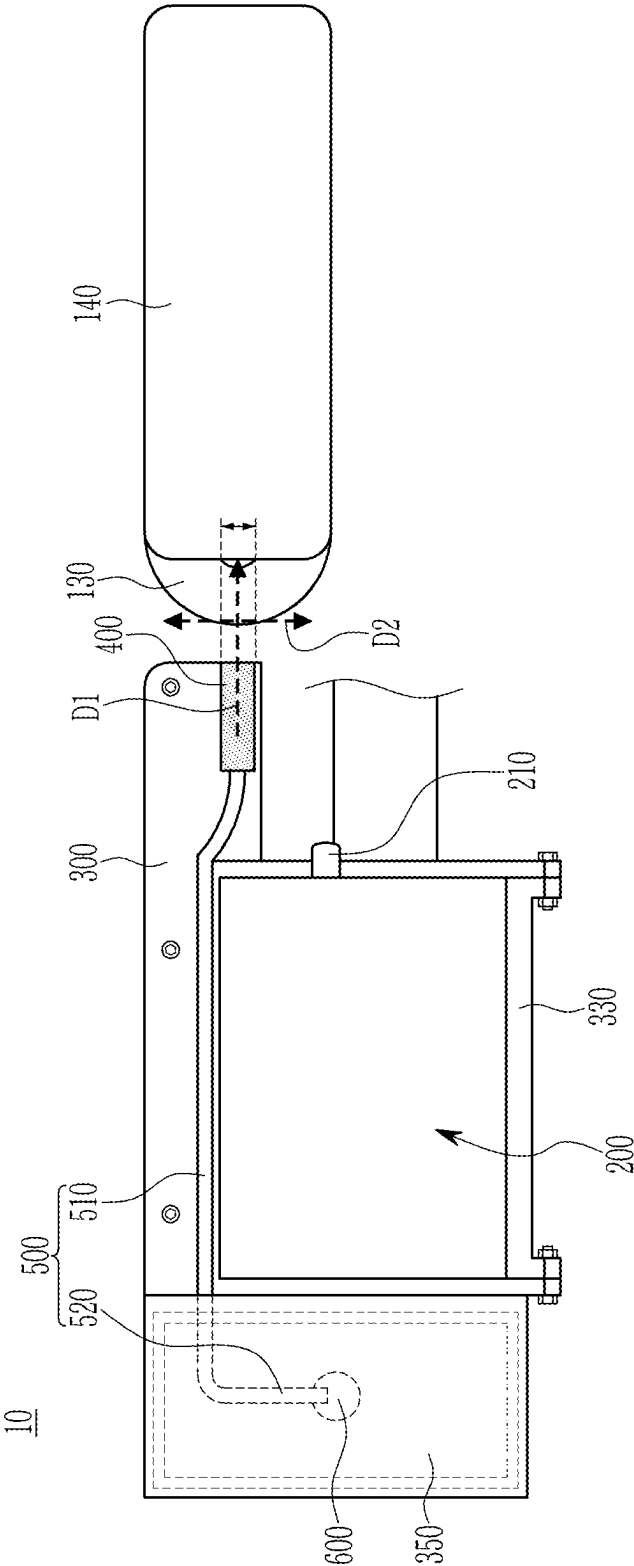


FIG. 10

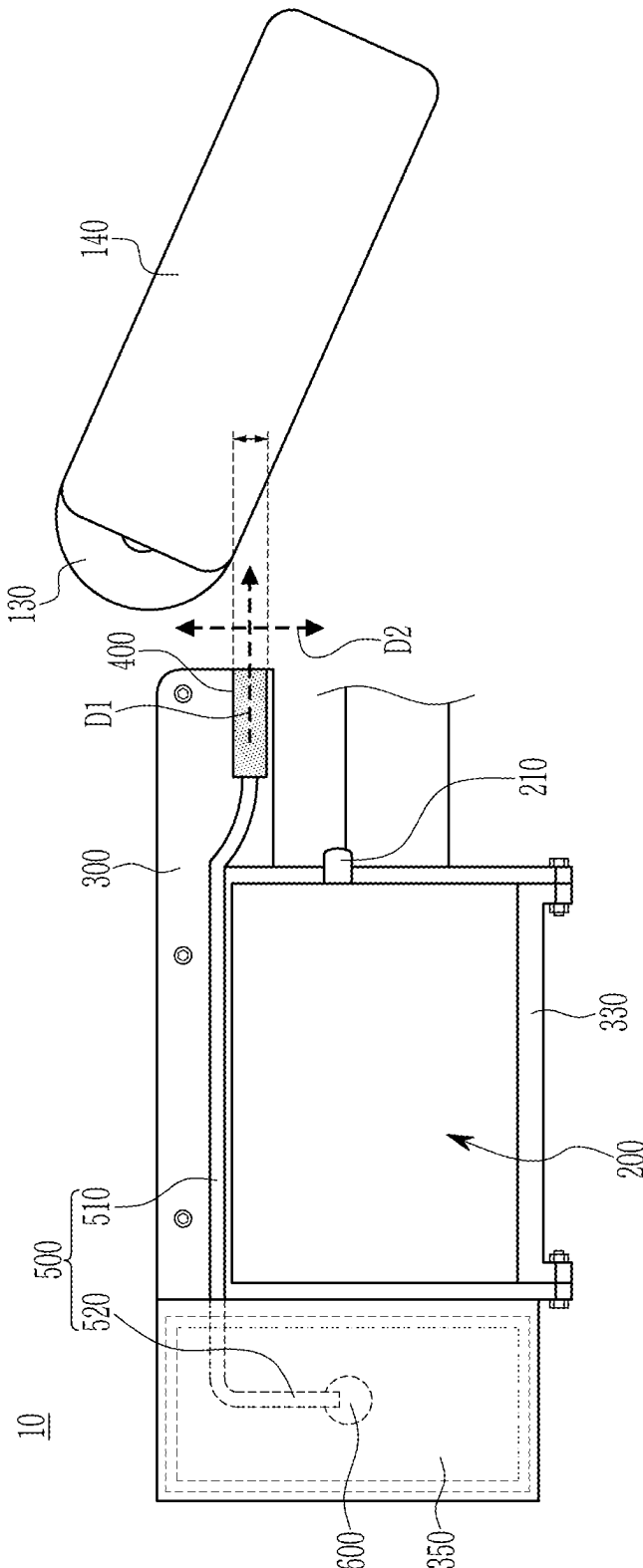
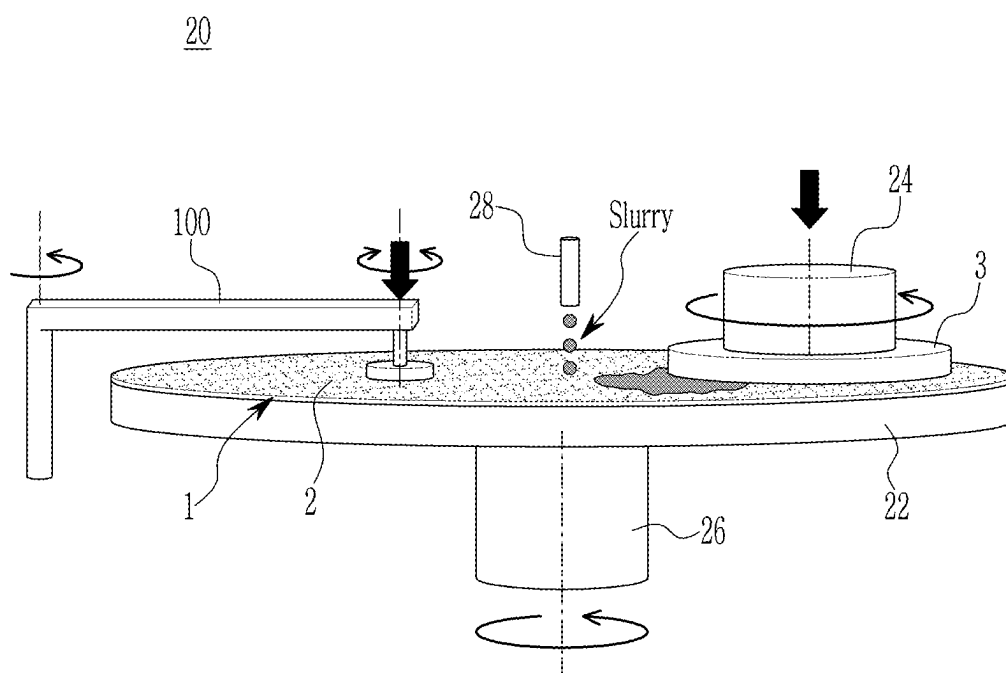


FIG. 11



**POLISHING PAD CONDITIONER POSITION
DETECTION MODULE, POLISHING
DEVICE, AND POLISHING DETECTION
METHOD USING THE POSITION
DETECTION MODULE**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0034061 filed in the Korean Intellectual Property Office on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(a) Field of the Invention

[0002] The present disclosure relates to a polishing pad conditioner position detection module, a polishing device, and a position detection method using the position detection module.

(b) Description of the Related Art

[0003] The chemical mechanical polishing or planarization CMP process of semiconductors is a process of flattening the wafer surface using chemical reactions and mechanical forces. It is a precise process in which not only mechanical factors such as the rotational speed of a polishing pad and a wafer, the pressure applied to the wafer, and the pattern directionality of the pad, but also chemical influences such as the interaction between slurry abrasive particles and a wafer surface and the role of the slurry organic additives are important variables.

[0004] On the point of polishing quality by the CMP process, the distribution of abrasive particles that are spread and maintained throughout the entire CMP polishing pad is important. In order to secure the performance of the CMP polishing pad, a process of conditioning the CMP polishing pad using a polishing pad conditioner is often required, and the conditioning process may include conditioning the upper part of the CMP polishing pad by placing the conditioner disk mounted on the polishing pad conditioner facing the CMP polishing pad.

[0005] However, in the process of conditioning the CMP polishing pad, there are instances where the conditioner disk mounted on the conditioner's holder may detach from the disk holder and also, during the movement of the conditioner arm equipped with a conditioner disk, issues such as degradation in the connecting components used to move the conditioner arm may occur, leading to errors in the movement position of the conditioner arm.

[0006] Accordingly, as the conditioner disk that is not placed in correct place on the CMP polishing pad conditions the CMP polishing pad there is a problem that uneven wearing occurs in the CMP polishing pad.

SUMMARY OF THE INVENTION

[0007] The present disclosure addresses some of these issues, and provides a polishing pad conditioner position detection module, a polishing device, and a position detection method using a position detection module, which can help prevent the deterioration of the polishing pad during the conditioning process by determining whether the disk is

mounted on the holder of the conditioner arm and whether the conditioner arm is in correct position through a detector fixed to the cleaning module for cleaning the disk.

[0008] According to an example embodiment, a polishing device includes a polishing pad conditioner position detection module, a polishing pad conditioner, a disk for conditioning a polishing surface on a polishing pad, and a cleaning module for cleaning the disk and placed adjacent to the polishing pad. The polishing pad conditioner position detection module includes a frame configured to surround the cleaning module; a housing connected to the frame; and a detector disposed inside the housing and configured to face a side of the disk when the disk is attached to the polishing pad conditioner. The detector is configured to detect whether the disk is mounted on the polishing pad conditioner and when mounted, whether the disk is in a correct predetermined position.

[0009] According to an embodiment, a polishing device includes a rotatable platen; a polishing pad placed on an upper surface of the platen and having a polishing surface; a polishing head disposed above the polishing pad and configured for mounting a substrate on a bottom surface of the polishing head; a driver configured to rotate the platen with respect to an axis perpendicular to the polishing surface of the polishing pad; a slurry supply outlet configured to supply slurry to the polishing surface of the polishing pad; a polishing pad conditioner for conditioning the polishing surface of the polishing pad; and a polishing pad conditioner position detection module configured to detect whether a disk of the polishing pad conditioner is correctly aligned.

[0010] According to an embodiment, a polishing pad conditioner position detection method for detecting whether a polishing pad conditioner including a conditioner arm, a head connected to the conditioner arm, and a disk mounted on the head is in correct position, includes: placing the polishing pad conditioner in cup position adjacent to a cleaning module for cleaning the disk; and detecting, using a detector placed on the cleaning module, whether a center of the disk is located within a predetermined distance in a first direction from the detector.

[0011] According to embodiments, when the polishing pad conditioner is placed in a cup position close to the cleaning module that cleans the conditioner disk, the detection part connected to the cleaning module detects whether the disk is mounted on the conditioner and whether there is a problem with the movement of the conditioner arm, thereby improving the thickness quality of the polishing pad by preventing the wearing down of the polishing pad during the conditioning process.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.

[0013] FIG. 2 is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.

[0014] FIG. 3 is a diagram illustrating a cross-section of a polishing pad conditioner position detection module according to an embodiment.

[0015] FIG. 4 is a diagram illustrating an exploded state for describing each configuration of a polishing pad conditioner position detection module according to an embodiment.

[0016] FIG. 5 is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.

[0017] FIGS. 6A-6C are diagrams for describing a polishing pad conditioner position detection module and method according to an embodiment.

[0018] FIG. 7 is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the disk is separated.

[0019] FIG. 8 is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the disk is separated.

[0020] FIG. 9 is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the conditioner is in correct position.

[0021] FIG. 10 is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the conditioner is in a correct position.

[0022] FIG. 11 is a view for describing a polishing device according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0023] Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the accompanying drawings so that those with ordinary skilled in the technical field to which the present disclosure belongs can easily implement it. The present disclosure may be implemented in various different forms and is not limited to the embodiments described herein.

[0024] In order to clearly describe the present disclosure in the drawings, certain parts which may be known in the art are omitted, and the same reference numerals are added to the same or similar components throughout the specification.

[0025] In addition, the size and thickness of each component shown in the drawing are arbitrarily expressed for convenience of explanation, so this disclosure is not necessarily limited to the illustration. In the drawings, for convenience of explanation, the thickness of layers, films, panels, regions, etc., are exaggerated for clarity.

[0026] Throughout the specification, it will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element (or using any form of the word “contact”), there are no intervening elements present at the point of contact. In addition, unless explicitly described to the contrary, the word “comprise”, and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of stated elements but not the exclusion of any other elements. Furthermore, when a component is described as “including” a particular element or group of elements, it is to be understood that the component is formed of only the element or the group of elements, or the element or group of elements may be combined with additional elements to form the component, unless the context indi-

cates otherwise. The term “consisting of,” on the other hand, indicates that a component is formed only of the element(s) listed.

[0027] In addition, when a part of a layer, film, region, plate, etc. is “above” or “on” another part, the reference part may be located above or below the reference part, depending on orientation, and does not necessarily mean “above” or “on” in the opposite direction of gravity.

[0028] In addition, throughout the specification, a plan view refers to a view from above, and a cross section view refers to a view from the side, unless the context indicates otherwise.

[0029] Ordinal numbers such as “first,” “second,” “third,” etc. may be used simply as labels of certain elements, steps, etc., to distinguish such elements, steps, etc. from one another. Terms that are not described using “first,” “second,” etc., in the specification, may still be referred to as “first” or “second” in a claim. In addition, a term that is referenced with a particular ordinal number (e.g., “first”) in a particular claim may be described elsewhere with a different ordinal number (e.g., “second”) in the specification or another claim.

[0030] Also these spatially relative terms such as “above” and “below” as used herein have their ordinary broad meanings—for example element A can be above element B even if when looking down on the two elements there is no overlap between them (just as something in the sky is generally above something on the ground, even if it is not directly above).

[0031] In the process of conditioning the CMP polishing pad of a CMP device, the conditioner disk mounted on the holder of the conditioner sometimes may deviate from the conditioner holder. Therefore, as the conditioning process proceeds on the CMP polishing pad without the conditioner disk being properly mounted on the holder, the quality of the CMP polishing pad may deteriorate.

[0032] In addition, in the process of moving the conditioner disk to face the CMP polishing pad while the conditioner arm is equipped with the conditioner disk, there are cases where the conditioner arm is moved to a position different from the target position due to problems such as deterioration of the parts related to the above-mentioned movement. In this case, during the conditioning process, the conditioner disk mounted on the conditioner arm is placed to be biased to one side on the CMP polishing pad, causing uneven abrasion of the CMP polishing pad.

[0033] The polishing pad conditioner position detection module 10 according to the present disclosure is configured to detect whether the detection part 400 is equipped with the disk 130 in the conditioner 100 and whether there is a problem with the movement of the conditioner arm 140.

[0034] Specifically, the detection part 400 is configured to determine whether the disk 130 is detached by checking whether the disk 130 is mounted on the polishing pad conditioner 100. For example, while the polishing pad conditioner position detection module 10 is placed in the cup position (e.g., a position where it is above a stationary cup that collects residue from a cleaning process), by checking whether the conditioner arm 140 is placed in a correct position and whether the disk 130 is detected within a predetermined range, it may be possible to determine whether there is an error in the movement of the conditioner arm 140.

[0035] According to some embodiments, the polishing pad conditioner position detection module 10 according to this disclosure first determines whether the disk 130 is mounted and positioned correctly before conditioning, and allows conditioning only if there are no issues, thereby preventing the polishing pad 1 from uneven abrasion during the conditioning process and thereby improving the thickness quality of the polishing pad 1.

[0036] Hereinafter, with reference to the drawings, a polishing pad conditioner position detection module 10, a polishing device 20, and a detection method using the position detection module according to an embodiment of the present disclosure will be described in more detail.

[0037] FIG. 1 is a diagram illustrating a polishing pad conditioner position detection module according to an embodiment, in which a head 110 and a disk 130 of the polishing pad conditioner 100 are shown in cup position adjacent to a cleaning module 200.

[0038] The cup position refers to the position where the disk 130 of the polishing pad conditioner 100 is placed when it is not being used to condition the top of the polishing pad 1 before conditioning the polishing surface 2.

[0039] For example, the cup position refers to a position in which the head 110 of the polishing pad conditioner 100 is located adjacent to the cleaning module 200 positioned next to the polishing pad 1 and is maintained in a standby mode before conditioning. The cup position may be a set standby position or a non-polishing position. For example, the cup position may refer to a predetermined set position in which the disk 130 and head 110 do not overlap the polishing pad 1 in a plan view. When the disk 130 is in the cup position, the cleaning module 200 may clean the disk 130 of the polishing pad conditioner 100.

[0040] As shown in FIG. 1, the polishing device 20 in this disclosure may include a rotatable platen 22, a polishing pad 1 disposed on the upper surface of the platen and having a polishing surface 2, a polishing head 24 disposed on the top of the polishing pad and mounting a substrate 3 on the bottom, a driving part 26 that rotates the platen 22 with respect to an axis perpendicular to the polishing pad 1, a slurry supplying part 28 (e.g., an outlet such as a nozzle connected to a slurry supply) that supplies slurry to the upper surface of the polishing pad 1, and a polishing pad conditioner 100 that conditions the polishing surface 2 of the polishing pad 1.

[0041] Referring to FIG. 1, in a polishing device 20 which includes a polishing pad conditioner 100 with a disk 130 for conditioning the polishing surface 2 and a cleaning module 200 positioned adjacent to the polishing pad 1 for cleaning the disk 130, the disclosed polishing pad conditioner position detection module 10 according to a present disclosure is configured to detect the position of the polishing pad conditioner 100 when the polishing pad conditioner 100 is in the cup position.

[0042] Specifically, the polishing pad conditioner position detection module 10 detects the position of the polishing pad conditioner 100 in the state where the polishing pad conditioner 100 is in the standby position. In some embodiments, the polishing pad conditioner 100 includes a holder 120 located under the head 110 of polishing pad conditioner 100. A disk 130 may be mounted on the holder 120 for conditioning the surface of the polishing pad 1. The disk 130 may be positioned near the cleaning module 200 for cleaning the disk 130 in a cup position.

[0043] As shown in FIG. 1, the head 110 is provided at the other end of the conditioner arm 140, and the disk 130 is mounted on the lower part of the head 110.

[0044] FIG. 1 shows a state in which the disk 130 is mounted on the head 110. Accordingly, the holder 120 disposed on the lower surface of the head 110 is omitted and does not appear in the drawing. The structure and arrangement position of the holder 120 can be seen in FIGS. 7 and 8.

[0045] The conditioning process is such that the disk 130 mounted on the conditioner arm 140 (specifically, the disk 130 mounted on the head 110) is placed to face the polishing pad 1, and the disk 130 is polished. Conditioning the upper part of the polishing pad 1.

[0046] First, the disk 130 presses the polishing pad 1 and rotates around the center of the disk 130 on an axis. Due to the pressure and rotation of the disk 130, friction occurs between the surface where the disk 130 and the polishing pad 1 come into contact. By the above friction, the polishing pad 1 is conditioned.

[0047] The process of the disk 130 pressing the polishing pad 1 includes the disk 130 descending toward the polishing pad 1, and the disk 130 moves until it contacts the polishing pad 1. Afterwards, the pressure applied to the disk 130 may be adjusted through a separate control unit depending on the degree of conditioning.

[0048] The process by which the disk 130 descends is as follows.

[0049] According to one embodiment, while the conditioner arm 140 and the head 110 connected to the conditioner arm 140 move up and down as a whole, the disk 130 mounted on the head 110 may move up and down. At this time, the drive shaft 150 supporting the conditioner arm 140 may have a structure whose length can be adjusted in the vertical direction, and the height at which the conditioner arm 140 is placed can be adjusted according to the length adjustment of the drive shaft 150.

[0050] As another example, the position (height) at which the conditioner arm 140 is disposed may be fixed. In this case, the head 110 connected to the conditioner arm 140 may have a structure capable of moving up and down.

[0051] Specifically, the head 110 is positioned in contact with the other end surface of the conditioner arm 140 (raised state, state in FIG. 1) and a position spaced a certain distance from the other end surface of the conditioner arm 140 (down state). The head 110 can move up and down. Here, when the head 110 is lowered, the disk 130 mounted on the head 110 is placed in contact with the polishing pad 1.

[0052] As another embodiment, the holder 120 provided on the lower surface of the head 110 may be moved up and down while maintaining the height of the conditioner arm 140 and the head 110. The process of the disk 130 applying pressure to the polishing pad 1 is a general process and may be similar to the process performed in an existing conditioning device. Information related to this is disclosed in Korean Patent Publication No. 10-2020-0043209, etc.

[0053] According to the present disclosure, the disk 130 is placed in contact with the holder 120 provided below the head 110. At this time, the holder 120 serves to couple the disk 130 to the lower part of the head 110. Referring to FIGS. 7 and 8, the holder 120 has a structure that protrudes from the lower part of the head 110.

[0054] According to an embodiment, the holder 120 may have a structure integrated with the head 110.

[0055] According to another embodiment, the length of the holder 120 protruding from the lower surface of the head 110 may be adjustable. At this time, the holder 120 may have a structure that allows it to move up and down. In this case, one side of the holder 120 may move up and down through the inside of the head 110.

[0056] The protruding portion of the holder 120 comes into contact with the upper surface of the disk 130 and serves to hold the disk 130. In other words, the holder 120 is coupled to the disk 130 and serves to couple the disk 130 to the head 110.

[0057] Referring to FIG. 8, the lower surface of the head 110 and the holder 120 have a step. Accordingly, the upper surface of the disk 130 may have a structure with a step so as to be in close contact with the step.

[0058] As shown in FIG. 1, the disk 130 may have a plate-shaped structure with a thickness. The disk 130 may be based on a metal base material and may have a structure including a polishing portion (a surface in contact with the polishing pad 1) where a plurality of diamond abrasive are adhered to the surface of the metal base material.

[0059] The disk 130 according to the present disclosure may have a structure including the step described above so that the opposite side of the polishing portion of the disk 130 is mounted on the holder 120 and the head 110.

[0060] The method of coupling the disk 130 and the holder 120 may be a magnetic attachment method using magnetic force, or a method using selective engagement of grooves and protrusions by rotation, etc. may be applied. The disk 130 may be coupled to the holder 120 in various ways and thus may be mounted on the head 110 provided with the holder 120.

[0061] Referring to FIGS. 7 and 8, the holder 120 may have a cylindrical shape. However, the structure of the holder 120 is not limited to the structure shown. The structure of the holder 120 is not limited as long as it is disposed on the lower surface of the head 110 and coupled to the upper surface of the disk 130.

[0062] As shown in FIG. 1, the head 110 may have a cylindrical shape. However, the shape of the head 110 is not limited to the shape shown in FIG. 1. The structure of the head 110 may vary depending on the structure of the holder 120 and the disk 130.

[0063] Although not shown in the drawing, the polishing pad conditioner 100 may further include a head driving unit that rotates the head 110. The head driving unit may rotate the head 110 based on the axis of the head 110.

[0064] Referring to FIG. 1, the head 110 can rotate based on the head 110 axis, and the holder 120 provided on the lower surface of the head 110 also rotates with the rotation of the head 110. At this time, the disk 130 coupled to the holder 120 and mounted on the lower part of the head 110 also rotates around the center of the disk 130.

[0065] In FIG. 1, as mentioned above, when the polishing pad conditioner position detection module 10 is placed in the cup position, the polishing pad 1 is not disposed below the disk 130 to overlap the disk 130 in the plan view.

[0066] In the cup position as shown in FIG. 1, the polishing pad conditioner position detection module 10 checks whether the disk 130 mounted on the polishing pad conditioner 100 is detected in correct position. Then to carry out the conditioning process, the polishing pad conditioner 100 is moved so that the disk 130 mounted on the conditioner arm 140 is positioned over the polishing surface 2 of the

polishing pad 1 as the conditioner arm 140 rotates around the drive shaft 150 (refer to FIG. 11).

[0067] As shown in FIG. 1, a polishing pad conditioner 100 may include a drive shaft 150 disposed in a direction perpendicular to a bottom surface of disk 130, a conditioner arm 140 having one end (a first end) connected to the drive shaft 150, a head 110 connected below the other end (a second end opposite the first end) of the conditioner arm 140, a holder 120 disposed on the lower surface of the head 110, and a disk 130 mounted on the holder 120 to condition the polishing surface 2 that is the upper surface of the polishing pad 1.

[0068] The polishing pad conditioner position detection module 10 according to embodiments of the present disclosure is connected to the cleaning module 200 and maybe positioned under the injection nozzle 210 of the cleaning module 200 as shown in FIG. 1. Here, the injection nozzle 210 of the cleaning module 200 serves to clean the disk 130 of the polishing pad conditioner 100, and the disk 130 may be cleaned by spraying liquid such as DIW (De-ionized Water) toward the disk 130 through the injection nozzle 210.

[0069] The injection nozzle 210 is disposed on one side of the cleaning module 200 facing the disk 130. In some embodiments, the cleaning module 200 may include an injection nozzle 210 for spraying DIW towards a side of the disk 130 and an injection part (which can be a second nozzle, not shown) disposed to spray DIW onto a lower surface of the disk 130 with the spray direction facing the lower surface of the disk 130. The injection part also serves to clean the disk 130.

[0070] Here, the side of the disk 130 means the outer periphery of the disk 130, which may include an outer sidewall of the disk 130, the portion mounted on the holder 120 is defined as the upper surface of the disk 130, and the opposite surface facing the upper surface is defined as the lower surface of the disk 130.

[0071] The polishing pad conditioner position detection module 10 according to some embodiments of the present disclosure may include a housing 300 placed on the outer surface of the cleaning module 200 and a detection part 400 placed inside the housing 300 so that one end faces the side of the disk 130, and by checking whether the disk 130 is detected within a predetermined distance through the detection part 400, it is possible to check whether the disk 130 is mounted on the holder 120.

[0072] FIG. 1 shows an embodiment of the polishing pad conditioner position detection module 10, and the polishing pad conditioner position detection module 10 may include a housing 300 disposed to be connected to one surface (e.g., one side) of the cleaning module 200 and a detection part 400 disposed inside the housing 300, and the housing 300 may include a protruding portion in which the detection part 400 is disposed.

[0073] In FIGS. 1-5, the housing 300 is connected to one side of the cleaning module 200 and is placed to protrude from the cleaning module 200 in the direction in which the disk 130 is located.

[0074] During detection, a first end of the detection part 400 disposed inside the housing 300 faces the side of the disk 130, and the first end of the detection part 400 detects the disk 130, so that it is possible to check whether the disk 130 is properly mounted in the holder 120.

[0075] Although FIGS. 1-5 show a structure in which the housing 300 is connected to one surface of the cleaning

module 200, a particular manner in which the housing 300 of the polishing pad conditioner position detection module 10 according to the present disclosure is fixed to the cleaning module 200 is not limited to this example.

[0076] In the drawings and embodiments to be described below, the housing 300 is placed on at least one side of the cleaning module 200 and further includes a supporting part 330 for fixing the housing 300 to the cleaning module 200. The supporting part 330 may surround all four side surfaces of the cleaning module 200.

[0077] The cable 500 connected to the detection part 400, which will be described below, is disposed inside the housing 300 and may have a structure electrically connected to the outside of the polishing pad conditioner position detection module 10 and cleaning module 200.

[0078] FIG. 2 is a diagram for describing a polishing pad conditioner position detection module according to an embodiment, FIG. 3 is a diagram illustrating a cross-section of a polishing pad conditioner position detection module according to an embodiment, and FIG. 4 is a diagram illustrating an exploded state for describing each configuration of a polishing pad conditioner position detection module according to an embodiment. FIG. 5 is a diagram for explaining a state in which the polishing pad conditioner position detection module 10 is coupled to the cleaning module 200.

[0079] First, FIG. 2 is a diagram showing an example perspective view of the polishing pad conditioner position detection module 10, and FIG. 3 is a diagram showing a cross-section seen from above of the polishing pad conditioner position detection module 10. Through FIG. 3, an example arrangement structure of the detection part 400 and the cable 500 disposed inside the polishing pad conditioner position detection module 10 are depicted.

[0080] FIG. 4 is an exploded view of the polishing pad conditioner position detection module 10, and it is possible to confirm the structure in which each component of the polishing pad conditioner position detection module 10 is combined.

[0081] As shown in FIG. 2 to FIG. 4, the polishing pad conditioner position detection module 10 according to the present disclosure includes a housing 300 placed on the outer surface of the cleaning module 200, a detection part 400 placed inside the housing 300 so that one end faces the side of the disk 130, and a supporting part 330 that fixes the housing 300 to the cleaning module 200, and the detection part 400 detects whether the disk 130 is mounted in the holder 120 of the polishing pad conditioner 100.

[0082] The supporting part 330 may be a support frame that is disposed to surround the periphery of the cleaning module 200 and that supports the housing. At least one surface of the supporting part 330, such as an outer surface of a side beam, is coupled to the housing 300 to fix the housing 300 to the cleaning module 200. The supporting part 330 may be a frame including four sides, each side being formed by a beam (e.g., a side beam). As depicted in FIGS. 2-4, some of the beams may be integrally formed and/or permanently affixed to each other, and at least one beam may be detachably connected to the other beams, for example using screws or a nut and bolt system or other attachment components. The beams may form a rectangular shape having inner side surfaces and outer side surfaces. The

beams that form the supporting part 330 may have a rigid structure and may be formed of a hard, rigid material such as a hard plastic.

[0083] Due to the rigid structure and the surrounding frame, the polishing pad conditioner position detection module 10 fixed by the supporting part 330 may be constructed so that it does not shake or so it minimizes any shaking even with external vibration, so that when the detection part 400 detects the position of the disk 130 and the conditioner arm 140, an error caused by the shaking of the polishing pad conditioner position detection module 10 may be minimized.

[0084] In some embodiments, the housing 300 may be formed of two joined bars or plates that have hollowed-out sections between them which house certain components. For example, inside the housing 300, a detection part 400 and a cable 500 connected to the detection part 400 may be disposed.

[0085] The detection part 400, which may be a detector such as a sensor, may be disposed inside a compartment formed in the housing 300 so that when the polishing pad conditioner position detection module 10 is positioned to be adjacent to the head 110 and disk 130, one end (e.g., a first end) faces the side surface of the disk 130, and the cable 500 may be connected to another end (e.g., a second, opposite end) of the detection part 400 to be disposed inside the housing 300.

[0086] The cable 500 has one end (e.g., a first end) connected to the second end of the detection part 400 and another (e.g., a second, opposite end) end extending to the outside of the housing 300, and serves to electrically connect the detection part 400 to the outside of the housing 300.

[0087] The cable 500 may include a first cable portion 510 connected to the second end of the detection part 400 and placed inside the housing 300, and a second cable portion 520 connected to the first cable portion 510 and extending outside the housing 300.

[0088] The polishing pad conditioner position detection module 10 according to the present disclosure may further include a receiving part 350 connected to the housing 300 to accommodate the cable 500. The receiving part 350 may further include a hole 600 for routing the second cable portion 520 externally.

[0089] As shown in FIGS. 2 to 4, one surface of the receiving part 350 (e.g., a lower outer surface facing the supporting part 330) may be coupled to the supporting part 330, and another surface of the receiving part 350 (e.g., a bottom surface of a bottom panel of the receiving part 350) may separately be coupled to the housing 300. Though not shown, the receiving part 350, which may be a compartment that houses the cable, may be connected to the supporting part 330 using attachment components such as screws or a nut and bolt system, and may be connected to the housing 300 using attachment components such as screws or a nut and bolt system. The housing 300, and the receiving part 350 may also be fixed to the cleaning module 200, as can be seen, for example, in FIG. 5, which depicts the supporting part 330 surrounding the cleaning module 200. The supporting part 330 (e.g., frame) may be affixed to the cleaning module 200 by surrounding outer surfaces of the cleaning module 200 and being tightened in place to securely attach to the cleaning module 200. In one embodiment, connector 250, which may be a connecting beam, connecting rod, or other rigid bar or arm-type structure, may connect the

cleaning module 200 to the drive shaft 150, etc. For example, part of the drive shaft 150 may be stationary and non-rotating (or may be rotatable separately from a portion that moves the conditioner arm 140), and may be affixed to a beam, rod, bar, or arm that connects to the cleaning module 200.

[0090] The detection part 400 and the cable 500 may be protected by being disposed inside the housing 300 and the receiving part 350.

[0091] An injection nozzle 210 may be attached on an outer surface of the cleaning module 200 and may protrude beyond an outer surface of the cleaning module 200. For example, the cleaning module 200 may have a parallelepiped shape, for example, in the form of a box having six sides, and may house cleaning pipes that transport cleaning solution to the injection nozzle 210. The cleaning solution may be received from outside the cleaning module 200 through piping, or may be included in the cleaning module 200, and spraying of the cleaning solution through the nozzle 210 may be controlled by a controller. The cleaning module 200 may be a spray dispenser. The detection part 400 may have a structure that protrudes from the cleaning module 200 more than the injection nozzle 210 in the same direction as the direction in which the injection nozzle 210 of the cleaning module 200 protrudes. The greater protruding amount of the protruding detection part 400 may result in the protruding detection part 400 being disposed closer to the disk 130 when the disk 130 is disposed adjacent to the cleaning module 200.

[0092] In this case, the housing 300 in which the detection part 400 is disposed also has a protruding structure together with the detection part 400, and the detection part 400 is disposed inside the housing 300. Therefore, the detection part 400 and the housing 300 in which the detection part 400 is disposed may be placed to protrude from the cleaning module 200 toward the disk 130.

[0093] This arrangement may prevent the DIW from also sticking to the detection part 400 located below the injection nozzle 210 in the process of spraying the DIW to clean the disk 130 in the injection nozzle 210. For example, by having a structure in which the housing 300 surrounds the protruding detection part 400, it is possible to protect the detection part 400 from DIW.

[0094] As shown in FIG. 4, the housing 300 may include a first housing portion 310 in which a detection part 400 and a cable 500 are disposed on its upper surface and a second housing portion 320 coupled to the top of the first housing portion 310. According to an embodiment, the first housing portion 310 and the second housing portion 320 may be coupled using screws, or nuts and bolts. Each of the first housing portion 310 and second housing portion 320 may be a plate, beam, or bar. Though shown to have a flat top surface, second housing portion 320 is not limited to this structure and shape and can have other structures or shapes. Similarly, though shown to have a flat bottom surface, first housing portion 310 is not limited to this structure and shape and can have other structures or shapes. The first housing portion 310 may be a first casing, or lower casing. The second housing portion 320 may be a second casing, or upper casing.

[0095] The first housing portion 310 may further include a guide part 340 for fixing the detection part 400 disposed therein.

[0096] Depending on the embodiment, the guide part 340 may also fix the first cable portion 510 connected to the detection part 400. The guide part 340 may have a groove structure that follows the shape of the detection part 400 and the first cable portion 510. The second housing portion 320 may also include a guide part 340. When the first housing portion 310 and second housing portion 320 are connected together, the guide parts 340 together may form a compartment with the housing 300 through which the detection part 400 and first cable portion 510 pass.

[0097] FIG. 5 is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.

[0098] FIG. 5 is a diagram for explaining a state in which the polishing pad conditioner position detection module 10 is coupled to the cleaning module 200, and the second housing portion 320 is coupled to the top of the first housing portion 310 in a state in which the first housing portion 310 and the supporting part 330 are coupled to the cleaning module 200.

[0099] In one embodiment, a lower portion of the receiving part 350 (e.g., a lower case portion) is fixed to the cleaning module 200 in a state of being coupled to the housing 300 and the supporting part 330, and an upper portion of the receiving part 350 (e.g., an upper case portion) may be coupled to overlap with the lower portion and fixed together.

[0100] Also, as shown in FIG. 5, the detection part 400 may be disposed close to the disk 130. The reason why the detection part 400 is disposed close to the disk 130 is to detect whether the target object, i.e., the disk 130, is located within a predetermined distance. The detection part 400 may be disposed close to the disk 130 based on, for example, the conditioner arm 140 being rotated to be adjacent to the detection part 400.

[0101] According to various embodiments, during detection by the detection part 400, no other object is positioned between the detection part 400 and the disk 130. Assuming that another object is located between the detection part 400 and the disk 130, the detection part 400 may incorrectly determine the object detected by the detection part 400 as the disk 130.

[0102] The detection part 400 may be a position detection sensor and may detect whether the disk 130 is located within a preset distance. For example, the detection part 400 may be a detector such as laser sensor, an image sensor, or another type of sensor that can sense or determine depth and/or distance.

[0103] When one end of the detection part 400 is arranged to face the direction in which the center of the disk 130 is located, if the direction is set to the first direction D1, the detection part 400 may detect whether the disk 130 is located within a predetermined distance in the first direction D1.

[0104] According to an embodiment, the detection part 400 may detect whether the disk 130 is located within a particular distance, such as 5 mm, from one end of the protruding detection part 400.

[0105] In the state where the polishing pad conditioner 100 which includes a holder 120 disposed on the lower surface of the head 110 and a disk 130 mounted on the holder 120 to condition the upper surface of the polishing pad 1 is positioned in a cup position close to the cleaning module 200 that cleans the disk 130, the polishing pad conditioner position detection module 10 according to the present dis-

closure may detect the position of part of the polishing pad conditioner 100. The polishing pad conditioner position detection module 10 may include a housing 300 disposed on an outer surface of the cleaning module 200 and a detection part 400 disposed inside the housing 300 and oriented toward a first direction D1 in which the center of the disk 130 is located.

[0106] The detection part 400 may detect whether a disk 130 is located within a predetermined distance in the first direction D1, and at the same time, whether a disk 130 is located within a predetermined width in the second direction D2 perpendicular to the first direction D1.

[0107] With regard to determining the predetermined width in the second direction D2, the detection part 400 may determine whether the disk 130 is located in a predetermined position having a particular width. When the disk 130 is not detected in that position to cover the particular width, the disk 130 may be detected as being misaligned.

[0108] Referring to FIG. 9, which will be described below, a detailed numerical description will be given. In FIG. 9, large arrows and small arrows are shown in the D2 direction, respectively. Please note that the large arrow is drawn to indicate that the distance is 5 mm in a straight line in the D1 direction. Note that the small arrow is drawn to indicate a gap of 3 mm in the D2 direction.

[0109] In the case of FIG. 9, the disk 130 is detected within a certain distance (e.g., 5 mm) in the D1 direction.

[0110] It is assumed that the shortest distance (straight line distance in the D1 direction) from the sensing unit 400 to the disk 130 is 5 mm. In FIG. 9, a large arrow in the D2 direction is shown so as to contact the circumference of the disk 130. The maximum protruding surface of the sensing unit 400 and the large arrow in the D2 direction are arranged horizontally, and the vertical distance across the two horizontal straight lines is 5 mm.

[0111] The case of FIG. 9 corresponds to a case where the disk 130 is detected over a certain width in the D2 direction.

[0112] According to the above example, the width of the disk 130 (width in the D2 direction) is detected within the 5 mm distance in the D1 direction described above. Therefore, the disk 130 does not need to be detected at a position deviating from a distance of 5 mm in the D1 direction.

[0113] The constant width in the D2 direction, as shown in FIG. 9, means the interval (width) indicated by the smaller arrow among the arrows shown in the D2 direction. Here, according to one example embodiment, it is assumed that the distance pointed by the small arrow shown in the D2 direction is 3 mm.

[0114] When referring to the above description, the disk 130 shown in FIG. 9 is detected within 5 mm in the D1 direction, and when looking at the disk 130 at the exact position spaced by 5 mm in the D1 direction, (that is, when looking at the disk 130 at the position where the large arrow in the D2 direction is shown), at that position, the disk 130 is detected in a 3 mm width range in the D2 direction.

[0115] For example, the polishing device 20 may be configured such that when the disk 130 is properly installed, and when the conditioner arm 140 is in a particular position, then the distance between the detection part 400 and the disk 130 in the first direction D1 will be a known predetermined amount, and the disk will also be detected to have a known predetermined width in the second direction D2. Therefore, when the detection part 400 performs detection and the distance in the first direction D1 and/or the width in the

second direction D2 does not match these predetermined amounts, then a misalignment may be detected, and either the disk 130, the conditioner arm 140, or both may be out of proper alignment.

[0116] The movement of the conditioner arm 140 as well as the operation of the detection part 400, spraying of the cleaning liquid, and processing and comparison of various data to determine whether the disk 130 and conditioner arm 140 are properly positioned may be controlled by a controller. For example, the controller can include one or more of the following components: at least one central processing unit (CPU) configured to execute computer program instructions to perform various processes and methods, random access memory (RAM) and read only memory (ROM) configured to access and store data and information and computer program instructions, input/output (I/O) devices configured to provide input and/or output to the processing controller 1020 (e.g., keyboard, mouse, display, speakers, printers, modems, network cards, etc.), and storage media or other suitable type of memory (e.g., such as, for example, RAM, ROM, programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, flash drives, and any type of tangible and non-transitory storage medium) where data and/or instructions can be stored. In addition, the controller can include antennas, network interfaces that provide wireless and/or wire line digital and/or analog interface to one or more networks over one or more network connections (not shown), a power source that provides an appropriate alternating current (AC) or direct current (DC) to power one or more components of the controller, and a bus that allows communication among the various disclosed components of the controller.

[0117] FIG. 11 is a view for describing a polishing apparatus according to an embodiment.

[0118] As shown in FIG. 11, the polishing apparatus 20 in this example may include a rotatable platen 22, a polishing pad 1 disposed on the upper surface of the platen 22 and having a polishing surface 2, a polishing head 24 disposed on the top of the polishing pad 1 and mounting the substrate 3 on the bottom, a driving part 26 that rotates the platen 22 with respect to an axis perpendicular to the polishing pad 1, a slurry supplying part 28 that supplies slurry to the upper surface of the polishing pad 1, and a polishing pad conditioner 100 that conditions the polishing surface 2 of the polishing pad 1.

[0119] During a polishing operation, as depicted in FIG. 11, as the platen 22 with the polishing pad 1 disposed on the top rotates about its axis, the polishing pad 1 also rotates together, and the polishing head 24 also rotates about the axis shown in FIG. 11, so that the substrate 3 mounted below the polishing head 24 rotates at the same time as the platen.

[0120] As the substrate 3 in contact with the polishing surface 2 of the polishing pad 1 rotates at the same time as the polishing pad 1, the substrate 3 in contact with the polishing surface 2 is polished, and slurry may be supplied from the slurry supplying part 28 during the polishing process.

[0121] As described in FIGS. 1 to 10, the polishing pad conditioner 100 included in the polishing device 20 may include a conditioner arm 140 with one end connected to a drive shaft 150. The conditioner arm may extend in a

direction that is perpendicular to an axis around which the disk 130 is disposed. A head 110 may be connected below another end of the conditioner arm 140, a holder 120 may be placed on the lower surface of the head 110, and a disk 130 may be mounted on the holder 120 to condition the polishing surface 2 which is the upper surface of the polishing pad 1. The disk 130 may be rotated and operated at the same time as the polishing head 24 rotating, so that the polishing pad 1 can be conditioned while polishing occurs.

[0122] The polishing pad conditioner position detection method using the polishing pad conditioner position detection module 10 according to the present disclosure is a method of detecting whether the polishing pad conditioner 100 is in correct position using the polishing pad conditioner position detection module 10 while the polishing pad conditioner 100 is placed in a particular position adjacent to the cleaning module 200 that cleans the disk 130.

[0123] By performing the detection, it can be determined whether the disk 130 is misaligned or is separated from the rest of the polishing pad conditioner 100 and/or whether there is an error in the movement of the conditioner arm 140.

[0124] Here, if the conditioner arm 140 is misaligned, for example, due to deterioration of the internal bearings disposed between the conditioner arm 140 and the driving shaft 150, this may mean that the degree to which the conditioner arm 140 is rotated by the rotation of the driving shaft 150 is not as much as the targeted position or is rotated beyond the targeted position.

[0125] A polishing pad conditioner position detection method using the position detection module 10 according to the present disclosure, as described above, first determines whether the disk 130 is mounted in the holder 120 and positioned in correct position prior to conditioning. If there is no problem (for example, if the disk 130 is mounted in the holder 120 and there is no error in the movement of the conditioner arm 140), by allowing the polishing pad conditioner 100 to condition the polishing pad 1, the thickness quality of the polishing pad 1 is improved by preventing uneven abrasion of the polishing pad 1 during the conditioning process.

[0126] The polishing pad conditioner 100 may have a structure including a conditioner arm 140 with one end connected to a driving shaft 150 and a second end connected to the disk 130, a head 110 connected below and affixed to the second end of the conditioner arm 140, a holder 120 placed on the lower surface of the head 110, and the disk 130 mounted on the holder 120 to condition the upper surface of the polishing pad 1. For information on the structure and coupling method of the head 110, the holder 120, and the disk 130 mounted thereon, refer to the above description.

[0127] The polishing pad conditioner position detection module 10 according to embodiments of the present disclosure is a structure fixed to the cleaning module 200, and when the polishing pad conditioner 100 is placed in a position close to the cleaning module 200, it is also located close to the polishing pad conditioner position detection module 10.

[0128] FIG. 6A is a diagram depicting various electronic and physical components of a polishing pad conditioner position detection module 10. FIGS. 6B, 6C, and 7-9 are diagrams for describing a method of detecting a position of the polishing pad conditioner 100.

[0129] For example, FIG. 7 and FIG. 8 are diagrams for describing a process in which a polishing pad conditioner

position detection module according to an embodiment detects whether the disk is separated.

[0130] FIG. 6A shows components of the polishing pad conditioner position detection module 10, such as for example, housing 300 (described previously), detection part 400 (described previously), decision part 700, which may include a computer software and/or hardware, which receives input information (e.g., distance, width, image information, etc.) relating to the sensed data received from the detection part 400, alarm part 800, which may include a computer software and/or hardware, and may additionally include visual and/or auditory generating components (e.g., a light or speaker) that issues an alarm if the detection part 400 detects an improperly positioned disk 130 (or a missing disk 130), and a control part 900, which may be or include the controller mentioned previously.

[0131] FIG. 6B illustrates a method of detecting a position of a polishing pad conditioner 100. In step S10, the polishing pad conditioner 100 is placed in a standby position adjacent to the cleaning module 200 for cleaning the disk 130. Next, in step S100, it is determined whether the disk 130 is positioned within a predetermined distance along a first direction D1 of the detection part 400 disposed on an outer surface of the cleaning module 200.

[0132] For example, FIG. 7 illustrates a state in which the disk 130 is mounted on the polishing pad conditioner 100 positioned at the standby position. In addition, FIG. 7 shows a state in which the conditioner arm 140 is in a correct position and one end of the detection part 400 of the polishing pad conditioner position detection module 10 is placed adjacent to and facing the disk 130.

[0133] However, if the conditioner arm 140 is not in the correct position and is tilted or otherwise misaligned, the disk 130 may not be located within a predetermined distance and the detection part 400 may detect a distance different from the predetermined distance.

[0134] Therefore, it is important to detect whether the conditioner arm 140 is in a normal position while the disk 130 is mounted on the polishing pad conditioner 100.

[0135] As shown in FIG. 7, the detection part 400 may be disposed to protrude from the cleaning module 200 toward the disk 130, and one end of the detection part 400 may be positioned adjacent to the disk 130. Within a predetermined distance indicated in FIG. 7, the detection part 400 may sense whether the disk 130 is properly connected.

[0136] FIG. 7 is a case where one end of the detection part 400 placed inside the housing 300 is placed toward the first direction D1 where the center of the disk 130 is located, and the disk 130 is located within a predetermined distance in the first direction D1. In this case, the detection part 400 may sense that the disk 130 is mounted on the holder 120 of the polishing pad conditioner 100 by the disk 130.

[0137] As shown in step S110 of FIG. 6B in connection with FIG. 6A, because the detected distance matches or is within a range of the predetermined distance in the first direction D1, decision part 700 determines that the holder 120 is equipped with a disk 130. It should be noted that decision part 700, alarm part 800, and control part 900 may all be portions of a computer program that controls performance of all of the steps discussed herein. Therefore, though described separately to distinguish the different steps, these parts may be separate software modules, or may be combined into a single program.

[0138] On the other hand, FIG. 8 shows a case where one end of the detection part 400 disposed inside the housing 300 is arranged to face the first direction D1 where the center of the disk 130 is supposed to be located, but the disk 130 may not be located within a predetermined distance in the first direction D1. For example, if the disk 130 is detached from the holder 120 of the polishing pad conditioner 100, the disk 130 may not be detected by the detection part 400 since there may be no detected object within a predetermined distance, so it is determined that the disk 130 is detached from the holder 120 of the polishing pad conditioner 100.

[0139] Therefore, as shown in FIGS. 6B and 8, when the disk 130 is not detected within a predetermined distance in the first direction D1, a step S120 in which the judgment part 700 determines that the disk 130 is detached from the holder 120 or is improperly attached to the holder occurs.

[0140] If the disk 130 is not detected within a predetermined distance in the first direction D1, in step S130 the alarm part 800 may generate an alarm.

[0141] In the method of detecting the position of the polishing pad conditioner 100 according to the present disclosure, in step S200, the detection part 400 detects whether the disk 130 is located within a predetermined distance in the first direction D1 and within a predetermined width in the second direction D2 perpendicular to the first direction D1.

[0142] The detection part 400 may detect whether the disk 130 is located within a predetermined distance in the first direction D1 from the detection part 400 and whether the disk 130 is located within a predetermined location and has a predetermined width in the second direction D2.

[0143] In the case where the disk 130 is detected to be located within the predetermined location having the predetermined width in the second direction D2, and the disk 130 is determined to be properly aligned. If the disk 130 is detected only in a partial area within the predetermined width, the disk 130 is determined to be improperly aligned.

[0144] FIG. 6C illustrates a method of detecting a position of a polishing pad conditioner 100. In step S10, the polishing pad conditioner 100 is placed in a standby position adjacent to the cleaning module 200 for cleaning the disk 130. Next, in step S200, it is determined whether the disk 130 is positioned within a predetermined distance in the first direction D1 and within a predetermined width in a second direction D2 perpendicular to the first direction.

[0145] As shown in FIG. 60, when the disk 130 is detected within a predetermined distance in the first direction D1 and within a predetermined width in a second direction D2 perpendicular to the first direction, it is determined that the disk 30 and the conditioner arm 140 are placed in a proper position (S210). In this case, the polishing pad conditioner 100 may perform conditioning (S220).

[0146] Conversely, if the disk 130 is not detected within a predetermined distance in the first direction D1 or the disk 130 is not detected within a predetermined width in a second direction D2, the disk 130 or the conditioner arm 140 is not placed at a proper position (S230). In this case, the alarm part 800 may generate an alarm (S240).

[0147] FIG. 9 and FIG. 10 are diagrams for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the conditioner is in a correct position.

[0148] FIG. 9 shows the case where the conditioner arm 140 is placed in correct position, and the polishing pad

conditioner position detection module 10 may detect the disk 130 within a predetermined distance in the first direction D1 and within a predetermined width in the second direction D2 perpendicular to the first direction D1. That is, the disk 130 is detected within a predetermined distance in the first direction D1 and within a predetermined width in the second direction D2.

[0149] Depending on the embodiment, a predetermined distance in the first direction D1 may be 5 mm, and a predetermined width in the second direction D2 may be 10 mm.

[0150] FIG. 9 and FIG. 10 are diagrams that omit the head 110 disposed between the conditioner arm 140 and the disk 130 for the purpose of illustration.

[0151] When the disk 130 is detected throughout the entire width in the second direction D2 as shown in FIG. 9, a step S210 in which the judgment part 700 determines that the conditioner arm 140 is in correct position occurs.

[0152] When the judgment part 700 determines that the conditioner arm 140 is in correct position, a step S220 in which the control part 900 adjusts the polishing pad conditioner 100 to condition the polishing pad 1 may be included.

[0153] As described above, in order for the polishing pad conditioner 100 to condition the polishing pad 1, it is first necessary to move the position of the polishing pad conditioner 100 from the cup position to a position in which the disk 130 contacts the upper surface of the polishing pad 1, for example, by the rotation of the driving shaft 150.

[0154] FIG. 10 shows a case where the conditioner arm 140 is not placed in correct position (step S230), and the polishing pad conditioner position detection module 10 cannot detect the disk 130 within a predetermined distance in the first direction D1 and/or within a predetermined width in the second direction D2 perpendicular to the first direction D1. FIG. 10 in particular shows a case where the disk 130 is not detected at all within a predetermined width in the second direction D2.

[0155] As shown in FIG. 10, the disk 130 may not be detected at all within a predetermined width in the second direction D2, but on the contrary, step S230 may occur in a case in which the disk 130 is detected only in an area that is only partially located within a predetermined width in the second direction D2.

[0156] As described above, when the disk 130 is detected only in a partial area, the detection part 400 does not detect the disk 130 as properly covering the width in the second direction D2. For example, in some embodiments, the disk 130 may only be detected in a partial area is because the conditioner arm 140 is tilted from its correct position.

[0157] When the disk 130 is not detected within a predetermined width in the second direction D2 as shown in FIG. 10, in step S230, the judgment part 700 determines that the position of the conditioner arm 140 is incorrect or misaligned.

[0158] In one embodiment, when the disk 130 is not detected within a predetermined width in the second direction D2, step S240 occurs, in which the alarm part 800 generates an alarm.

[0159] As described above, there should be no interference between the polishing pad conditioner position detection module 10 and other previously used CMP components in how the polishing pad conditioner position detection module 10 detects the position of the polishing pad conditioner 100.

[0160] Specifically, the polishing pad conditioner position detection module 10 should have no interference with the platen shield surrounding the platen 22 and there should be no interference with the injection nozzle 210 of the cleaning module 200 and the injection part (not shown) described above. In addition, there should be no collision with the polishing pad conditioner position detection module 10 while the polishing pad conditioner 100 rotates by the driving shaft 150.

[0161] In the conditioning process of the conventional polishing pad conditioner, the disk may be disengaged, or the disk may be placed in a position different from the target position on the CMP polishing pad due to problems such as deterioration of parts of the conditioner arm. As a result, there was a problem of conditioning the CMP polishing pad with the conditioner disk positioned off-center on the CMP polishing pad. In those cases, it would often be necessary to rely on human technicians to detect misalignment of the conditioner disk, and such detection might not occur until significant damage has been done to polishing pads and/or devices being manufactured.

[0162] The polishing pad conditioner position detection module 10 according to the present disclosure detects in advance whether the detection part 400 is in a state where the disk 130 is mounted on the conditioner 100 and whether there is a problem with the movement of the conditioner arm 140, thereby preventing the CMP polishing pad 1 from being worn unevenly and improving the quality of the CMP polishing pad 1.

[0163] In particular, the polishing pad conditioner position detection module 10 according to the present disclosure is an additional configuration that is combined with the cleaning module 200 for cleaning the polishing pad conditioner 100. Therefore, the polishing pad conditioner position detection module 10 according to the present disclosure may be highly useful in that it can be utilized while using the existing polishing pad conditioner 100 without any modification.

[0164] Although the embodiments of the present disclosure have been described above, the embodiments are not limited thereto, and it is possible to implement various modifications within the scope of the claims, detailed explanations, and accompanying drawings.

What is claimed is:

1. A polishing pad conditioner position detection module for a polishing device which includes a polishing pad conditioner, a disk for conditioning a polishing surface on a polishing pad, and a cleaning module for cleaning the disk and placed adjacent to the polishing pad, the polishing pad conditioner position detection module comprising:

- a frame configured to surround the cleaning module;
 - a housing connected to the frame; and
 - a detector disposed inside the housing and configured to face a side of the disk when the disk is attached to the polishing pad conditioner,
- wherein the detector is configured to detect whether the disk is mounted on the polishing pad conditioner and when mounted, whether the disk is in a correct predetermined position.

2. The polishing pad conditioner position detection module of claim 1, wherein:

- the detector is configured to detect whether the disk is mounted on the polishing pad conditioner and when mounted, whether the disk is in the correct predetermined position while a portion of polishing pad con-

ditioner configured to hold the disk is positioned in a standby position adjacent to the cleaning module.

3. The polishing pad conditioner position detection module of claim 1, wherein:

- the detector is positioned to detect whether the disk is in the correct predetermined position based on the disk being mounted on a holder located underside a head of the polishing pad conditioner.

4. The polishing pad conditioner position detection module of claim 1, wherein:

- one end of the detector is arranged to face a first direction toward a center of the disk when the disk is in the correct predetermined position and is configured to detect the disk positioned within a predetermined distance in the first direction.

5. The polishing pad conditioner position detection module of claim 4, wherein:

- the detector is configured to detect whether the disk is located within a predetermined location having a width in a second direction which is perpendicular to the first direction.

6. The polishing pad conditioner position detection module of claim 1, further comprising a cable connected to the detector and extending out of the housing.

7. The polishing pad conditioner position detection module of claim 6, wherein the cable comprises:

- a first cable portion connected to the detector and placed inside the housing; and
- a second cable portion connected to the first cable portion and extending out of the housing.

8. The polishing pad conditioner position detection module of claim 6, further comprising a receiving part connected to the housing for receiving the cable.

9. The polishing pad conditioner position detection module of claim 8, wherein the receiving part further comprises a hole for routing the cable externally.

10. The polishing pad conditioner position detection module of claim 1, wherein the housing is connected to outer surface of the cleaning module.

11. The polishing pad conditioner position detection module of claim 1, wherein the housing further comprises a guide part inside which the detector is placed.

12. The polishing pad conditioner position detection module of claim 1, wherein the detector and the housing in which the detector is placed are positioned to protrude from the cleaning module toward the disk.

13. A polishing device comprising:

- a rotatable platen;
- a polishing pad placed on an upper surface of the platen and having a polishing surface;
- a polishing head disposed above the polishing pad and configured for mounting a substrate on a bottom surface of the polishing head;
- a driver configured to rotate the platen with respect to an axis perpendicular to the polishing surface of the polishing pad;
- a slurry supply outlet configured to supply slurry to the polishing surface of the polishing pad;
- a polishing pad conditioner for conditioning the polishing surface of the polishing pad; and
- a polishing pad conditioner position detection module configured to detect whether a disk of the polishing pad conditioner is correctly aligned.

14. A polishing pad conditioner position detection method for detecting whether a polishing pad conditioner including a conditioner arm, a head connected to the conditioner arm, and a disk mounted on the head is in correct position, the method comprising:

placing the polishing pad conditioner in cup position adjacent to a cleaning module for cleaning the disk; and detecting, using a detector placed on the cleaning module, whether a center of the disk is located within a predetermined distance in a first direction from the detector.

15. The polishing pad conditioner position detection method of claim **14**, comprising:

detecting whether the disk is located within a predetermined distance in the first direction and within a predetermined width in a second direction perpendicular to the first direction.

16. The polishing pad conditioner position detection method of claim **15**, comprising:

determining that the disk is disposed on a conditioner arm when the disk is detected within a predetermined distance in the first direction; and

determining that the conditioner arm is in the correct position when the disk is detected within a predetermined width in the second direction.

17. The polishing pad conditioner position detection method of claim **16**, comprising:

determining that the disk is separated from a holder placed underside a head of the polishing pad conditioner when the disk is not detected within a predetermined distance in the first direction.

18. The polishing pad conditioner position detection method of claim **15**, comprising:

determining that a position of the conditioner arm is displaced when the disk is not detected within a predetermined width in the second direction.

19. The polishing pad conditioner position detection method of claim **14**, comprising:

generating an alarm when the disk is not detected within a predetermined distance in the first direction.

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